

Fathimathul Mina C H

AI Engineer | Machine Learning | Deep Learning | Generative AI

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SUMMARY

Results-driven AI Engineer with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Generative AI. Skilled in building intelligent systems—from real-time facial emotion detection to multi-agent LLM platforms and RAG-powered applications. Passionate about applying cutting-edge AI research to solve real-world problems, with a strong foundation in end-to-end model development, fine-tuning, and deployment.

TECHNICAL SKILLS

Languages: Python, SQL

AI/ML: Machine Learning, Deep Learning, CNNs, RNNs, Transformers, Computer Vision, NLP, Sentiment Analysis, Model Fine-Tuning (Wav2Vec2, BERT, ResNet), Generative AI, Reinforcement Learning basics

LLM Ecosystem: LangChain, GROQ, Llama 3.3 70B, Gemini API, RAG (Retrieval-Augmented Generation), LangGraph, Hugging Face

Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, PyTorch, OpenCV, Dlib, OpenFace, Pandas, NumPy, Matplotlib, FAISS

Web & APIs: Django REST Framework, FastAPI, Streamlit, REST APIs

Tools & Platforms: Git, GitHub, Docker, Jupyter Notebook

MAIN PROJECTS

RAGFlow – Intelligent Document Q&A with RAG Pipeline

[Live Demo](#)

Python | LangChain | FAISS | LLaMA | Streamlit | Hugging Face | RAG

A production-ready Retrieval-Augmented Generation (RAG) application enabling users to interact with documents through natural language, delivering precise, context-aware answers powered by LLM reasoning.

- **RAG Architecture:** Engineered an end-to-end RAG pipeline that ingests and chunks documents, creates semantic embeddings using FAISS vector store, and retrieves the most relevant context before LLM inference—dramatically reducing hallucinations.
- **LLM Integration:** Integrated LLaMA model via LangChain to generate grounded, citation-backed responses from retrieved document segments, achieving high factual accuracy.
- **Deployment:** Deployed the full application on Streamlit Cloud with a clean, interactive UI allowing multi-document upload, real-time querying, and conversation history.

Technologies: *Python | LangChain | FAISS | LLaMA | Streamlit | Hugging Face | Git*

AgentForge – Multi-Agent AI Collaboration Platform *[In Progress]*

[GitHub](#)

Python | LangChain | LangGraph | GROQ | Llama 3.3 70B | Next.js (Planned)

Currently building a sophisticated multi-agent AI platform enabling users to assemble teams of specialized AI agents (Researcher, Analyst, Writer, Critic) that collaboratively tackle complex tasks through intelligent orchestration.

- **Multi-Agent Architecture:** Designed a multi-agent orchestration framework with distinct agent roles and inter-agent communication, enabling collaborative problem decomposition and solution synthesis.
- **Core Agent System:** Built Phase 0 with a fully functional single-agent system integrated with web search tools using GROQ and Llama 3.3 70B, demonstrating real-time information retrieval and reasoning.
- **Roadmap & Scalability:** Architected a scalable roadmap featuring persistent memory, dynamic agent spawning, task-routing, and a full Next.js web interface for user-controlled agent team configuration.

Technologies: *Python | LangChain | LangGraph | GROQ | Llama 3.3 70B | Git | Next.js (Planned)*

Facial Emotion Recognition (FER) – Real-Time Emotion Detection System

[GitHub](#)

Python | TensorFlow | Keras | OpenCV | Dlib | OpenFace | CNN

A real-time facial emotion recognition system leveraging deep learning and computer vision to detect and classify human emotions from live video streams with high accuracy across 7 emotion categories.

- **Deep Learning Model:** Developed and trained a custom CNN model on the FER2013 dataset achieving strong validation accuracy across 7 emotions: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise.
- **Computer Vision Pipeline:** Implemented face detection pipeline using OpenCV Haar Cascade, Dlib landmark detection, and OpenFace for robust face alignment prior to emotion classification—ensuring stable real-time inference from webcam feeds.
- **Model Optimization:** Applied advanced fine-tuning techniques including data augmentation, dropout regularization, and batch normalization to improve generalization and reduce overfitting on the FER2013 dataset.

Technologies: *Python | TensorFlow | Keras | OpenCV | Dlib | OpenFace | CNN | FER2013*

OTHER PROJECTS

ML & Deep Learning Pipelines

[GitHub](#)

Python | TensorFlow | Keras | Scikit-learn

- Built end-to-end ML pipelines covering data preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation; deployed models for real-world predictions.

Technologies: *Python | TensorFlow | Keras | Scikit-learn | Pandas | NumPy*

Road Accidents India – Predictive Modeling

[GitHub](#)

Python | Scikit-learn | Pandas | Jupyter Notebook

- Built regression and classification models to predict road accident numbers and classify high-risk years across India (2016–2022); Linear Regression achieved $R^2 \approx 0.993$, Logistic Regression attained 100% accuracy. Included Kerala-focused analysis.

Technologies: *Python | Scikit-learn | Pandas | NumPy | Jupyter Notebook*

Expense Tracker API – RESTful Personal Finance Backend

[GitHub](#)

Python | Django REST Framework

- Developed a fully functional RESTful API for managing personal expenses with complete CRUD endpoints, authentication, and structured data models using Django REST Framework.

Technologies: *Python | Django REST Framework | SQLite | Git*

EDUCATION

Machine Learning Training

2025 – Present

Peckapeer Academy Private Limited (Brototype), Thalassery

M.Sc. Computer Science with Specialisation in AI

2023 – 2025

Government College Thalassery | Score: 63%

B.Sc. Computer Science

2020 – 2023

NAM College Kallikandy | Score: 74%